Sarah A. Carneiro, PhD.

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Professional Summary

AI researcher with a PhD in Signal, Image, and Automation and postdoctoral experience in multimodal AI, foundation models, biomolecular language generation, and LLM evaluation. Research experience spans contrastive learning, parameter-efficient fine-tuning (LoRA), benchmark design, human and model-assisted evaluation, spatio-temporal graph learning, and computer vision. Reported results include up to a 4-point improvement over state-of-the-art text-generation baselines across approximately 250K protein entries and a 78% reduction in MSE for road-speed prediction versus an MLP baseline. Published work includes a NeurIPS 2025 paper.

Core Skills

- **AI Research:** Multimodal representation learning, foundation models, LLMs, multimodal generation, contrastive learning, instruction tuning, LoRA, self-supervised learning, biomolecular AI
- **Evaluation:** Benchmark design, LLM benchmarking, ablation studies, semantic and factuality metrics, LLM-as-a-judge, expert human validation, robustness and generalization analysis
- **Deep Learning:** PyTorch, CNNs, GNNs, RNNs, temporal architectures, Transformer-based approaches, cross-modal alignment and fusion
- **Compute and Data:** CUDA, NVIDIA A100 GPUs, multi-GPU experimentation, Linux, SSH, Slurm, batch-job submission and monitoring, MongoDB, PyMongo, GPS trajectory preprocessing
- **Programming and Frameworks:** Python, Hugging Face Transformers, TensorFlow, OpenCV, SciPy, Java, Git

Research Experience

Postdoctoral Researcher — Multimodal AI and Foundation Models

École Polytechnique, LIX Paris, France | 2025–Present

- *Multimodal Scientific AI:* Develop research models integrating protein sequences, molecular structures, and natural language for scientific applications.
- *Model Training and Ablation Studies:* Investigate contrastive learning and compare full fine-tuning with LoRA, analyzing model performance and generation quality.
- *LLM Evaluation and Hallucination Assessment:* Build evaluation pipelines combining expert human validation and LLM-as-a-judge to assess correctness, relevance, and hallucinations across 351K protein–protein interaction pairs.
- *Benchmarking and Experimental Design:* Design experiments, preprocessing pipelines, and protein-captioning benchmarks, reporting improvements of up to 4 points over state-of-the-art baselines on text-generation metrics across approximately 250K protein entries.
- *Benchmark Leadership:* Lead development of a benchmark for fine-grained French cultural and regional knowledge; evaluate 32 LLMs (1B–30B parameters) on 3,132 human-validated questions across 13 regions.
- *LLM Benchmarking and Robustness:* Lead French-specific LLM benchmarking within OPENLLM France, including experimental design, ablation studies, and robustness analysis.
- *Prompt Engineering and Safeguards:* Design task-specific prompts and safeguards for generation, summarization, question filtering, and evaluation in scientific and domain-specific LLM applications.
- *Model Implementation and Compute:* Implement and train models using Python, PyTorch, CUDA, and NVIDIA A100 GPUs.
- *Research Strategy and Grant Development:* Propose and guide internship projects, formulate research hypotheses, contribute to research directions, develop grant proposals for Hi! PARIS and ANR initiatives, and engage with potential industry partners on applied research challenges.
- *Project and Research Coordination:* Coordinate project timelines, task allocation, manuscript submission workflows, and collaboration across research projects.

PhD Researcher — Structure-Aware Deep Learning for Spatio-Temporal Data

IFP Energies Nouvelles / Université Gustave Eiffel France | 2020–2024

- *Spatio-Temporal Traffic Prediction:* Designed a deep learning framework for speed prediction in data-sparse road networks, integrating road topology, road clustering, and sparse GPS trajectories.

- *Deep Learning Model Comparison:* Compared CNNs, Graph Neural Networks (GNNs), and Recurrent Neural Networks (RNNs), and explored Transformer-based approaches for spatio-temporal prediction.
- *Graph Representation and Efficiency:* Evaluated alternative graph representations to identify effective and computationally feasible strategies for road-network prediction.
- *Model Performance Improvement:* Reduced mean squared error by 78% compared with an MLP baseline (6.75 vs. 31.03) by incorporating road-infrastructure features.
- *Real-World Data Preprocessing:* Developed preprocessing pipelines for noisy GPS trajectories, including data imputation and smoothing.
- *Large-Scale Data Processing:* Retrieved and processed GPS trajectory data from MongoDB databases using PyMongo and conducted comparative experiments under data-sparse conditions.

Research Scientist — Deep Learning for Spatio-Temporal Video Understanding
Semantix Campinas, Brazil | 2018–2019

- *Video Classification and Pose Estimation:* Developed multi-stream convolutional neural network ensembles for video classification and human pose estimation.
- *Multimodal Feature Fusion:* Fused visual features including depth estimation, optical flow, and pose estimation; applied data-balancing strategies.
- *Ablation Studies and Error Analysis:* Conducted ablation studies on multimodal feature combinations and investigated false-positive detections through frame-level verification and error analysis.
- *Experimental Implementation and Evaluation:* Implemented and evaluated experiments on Linux-based AWS instances, collaborating with teammates to interpret results and refine experiments.

Selected Publications

- **Prot2Text-V2: Multimodal Protein-to-Text Generation for Function Prediction.** NeurIPS, 2025.
- **PPI2Text: Captioning Protein-Protein Interactions with Coordinate-Aligned Pair-Map Decoding.** arXiv preprint, 2026.
- **CARTE: A Benchmark for Mapping Language Model Knowledge Across France.** arXiv preprint, 2026.
- **The Masked Advantage: Uncovering Local-Language Access to Cultural Knowledge in LLMs.** arXiv preprint, 2026.
- **Every Wrong Answer Counts: Option-Level Psychometrics for LLM Multiple-Choice Benchmarks.** arXiv preprint, 2026.
- **SWMLP: Shared Weight Multilayer Perceptron for Car Trajectory Speed Prediction Using Road Topographical Features.** MT-ITS, 2023.
- **Graph-Based Supervoxel Computation from Iterative Spanning Forest.** International Conference on Discrete Geometry and Mathematical Morphology, 2021.
- **Fight Detection in Video Sequences Based on Multi-Stream Convolutional Neural Networks.** SIBGRAPI, 2019.

Research Leadership and Supervision

- *Research Direction and Problem Formulation:* Contribute to defining research directions, formulating technical questions, and guiding project development across AI research topics.
- *Agentic Systems and LLMs:* Co-supervise research on agentic systems and constrained language-model generation, guiding technical exploration and research development.
- *LLM Evaluation and Efficiency:* Supervise projects on LLM benchmarking and evaluation under quantization.
- *Applied AI and Computer Vision:* Supervise projects in medical image analysis, AI-generated and manipulated image detection, and plant disease classification.

Education

Ph.D. in Signal, Image, and Automation — Université Gustave Eiffel, France 2020–2024
M.Sc. in Computer Science — University of Campinas (UNICAMP), Brazil 2020

Certifications

- AI Literacy for Business Leaders — IBM, 2026.
- Generative AI with Large Language Models — AWS, 2024.

Languages

Portuguese | English | French